Multiplication and Inverse Matrices

Five ways to see a matrix product, all giving the same numbers. Then the inverse: when it exists, and how Gauss–Jordan finds it.

PART 1 Index

What this lesson covers, in order. Click a topic to jump to it.

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   Each entry of AB is a row of A times a column of B. Which shapes can be multiplied.
2. **By columns and by rows**
   Columns of AB mix the columns of A; rows of AB mix the rows of B. Many portfolios at once.
3. **Columns times rows**
   AB as a sum of simple pieces, one per fund.
4. **Block multiplication**
   Cut matrices into blocks and multiply the blocks.
5. **The inverse, and when it exists**
   $A^{-1}A = I$. No inverse when some trade changes nothing: $Ax = 0$ with $x \neq 0$.
6. **Gauss–Jordan: 2 × 2**
   Find A^{-1} by eliminating $[A \mid I]$ until it becomes $[I \mid A^{-1}]$.
7. **Gauss–Jordan: 3 × 3, and why it works**
   The three-fund portfolio, step by step. $E[A \mid I] = [I \mid E]$, so $E = A^{-1}$.
8. **Using the inverse**
   Every target at once, building-block trades, the 2 × 2 formula, and when not to compute A^{-1}.

METHODS IN THIS LESSON: Entry rule · Shapes rule · Column way · Row way · Column × row (outer products) · Block multiplication · Singular test (Ax = 0) · Gauss–Jordan elimination · 2 × 2 inverse formula

PART 2 Prerequisites

Click any link to review it, then come back.

- Matrix × vector, two ways (Lesson 1): Ax mixes the columns of A.
- Singular matrices (Lesson 1): funds that add no new direction.
- Elimination on the augmented matrix (Lesson 2): the portfolio eliminated to U.
- Elimination matrices and "row × matrix mixes rows" (Lesson 2).
- Undoing one step (Lesson 2): the inverse of a single elimination matrix.

PART 3 Explanation

THE RUNNING EXAMPLE: MANY CLIENTS, ONE SET OF FUNDS

Our funds, as in every lesson:

Fund	Return per ₹1 lakh	Market exposure (beta) per ₹1 lakh
Nifty 50 index fund	₹12 thousand (12%)	1.0
Government bond fund	₹7 thousand (7%)	0
Midcap fund	₹15 thousand (15%)	1.2

Until now we multiplied the fund matrix A by **one** portfolio at a time: $Ax = b$. A desk has **many** clients. So we stack their portfolios side by side, one client per column, into a matrix X. With two funds and two clients:

$$A = \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix} \begin{matrix} \leftarrow \text{budget used per ₹1 lakh} \\ \leftarrow \text{return earned per ₹1 lakh} \end{matrix} \qquad X = \begin{bmatrix} 7 & 4 \\ 3 & 6 \end{bmatrix} \begin{matrix} \leftarrow \text{₹ lakh in Nifty} \\ \leftarrow \text{₹ lakh in bonds} \end{matrix}$$

Column 1 of X is client 1: ₹7 lakh Nifty, ₹3 lakh bonds. **Column 2** is client 2: ₹4 lakh Nifty, ₹6 lakh bonds. The product AX will give every client's (budget, return) at once. The **inverse** A^{-1} will then do the reverse: targets in, portfolios out.

1. The entry rule and shapes

Start from what we already know. A dot product of two lists of the same length multiplies matching entries and adds them: $(1, 1) \cdot (7, 3) = 1 \times 7 + 1 \times 3 = 10$. Lesson 1 used it to compute Ax row by row. A matrix product is nothing more than many dot products, one for every (row, column) pair.

METHOD · ENTRY RULE (ROW TIMES COLUMN)

The entry in row i and column j of $C = AB$ is the dot product of **row i of A** with **column j of B**:

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} = \sum_{k=1}^{n} a_{ik}\, b_{kj}.$$

The index k walks along row i of A and, at the same time, down column j of B.

All four entries of AX, one at a time

1

WHAT WE DO — Find c_{11}, the entry in row 1, column 1.

HOW — Row 1 of A is $(1, 1)$. Column 1 of X is $(7, 3)$. Dot product: $1 \times 7 + 1 \times 3 = 7 + 3 = 10$.

MEANING — Row 1 of A is the budget rule, column 1 of X is client 1. So $c_{11} = 10$ is **client 1's budget**: ₹10 lakh.

2

WHAT WE DO — Find c_{12} (row 1, column 2).

HOW — Row 1 of A, $(1, 1)$, with column 2 of X, $(4, 6)$: $1 \times 4 + 1 \times 6 = 4 + 6 = 10$.

MEANING — **Client 2's budget**: also ₹10 lakh.

3

WHAT WE DO — Find c_{21} (row 2, column 1).

HOW — Row 2 of A, $(12, 7)$, with column 1 of X, $(7, 3)$: $12 \times 7 + 7 \times 3 = 84 + 21 = 105$.

MEANING — **Client 1's return**: ₹7 lakh in Nifty earns ₹84 thousand, ₹3 lakh in bonds earns ₹21 thousand, total ₹105 thousand.

4

WHAT WE DO — Find c_{22} (row 2, column 2).

HOW — Row 2 of A, $(12, 7)$, with column 2 of X, $(4, 6)$: $12 \times 4 + 7 \times 6 = 48 + 42 = 90$.

WHAT WE HAVE NOW — All four entries:

$$AX = \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix} \begin{bmatrix} 7 & 4 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 1 \cdot 7 + 1 \cdot 3 & 1 \cdot 4 + 1 \cdot 6 \\ 12 \cdot 7 + 7 \cdot 3 & 12 \cdot 4 + 7 \cdot 6 \end{bmatrix} = \begin{bmatrix} 10 & 10 \\ 105 & 90 \end{bmatrix} \begin{matrix} \leftarrow \text{budgets} \\ \leftarrow \text{returns} \end{matrix}$$

Reading the sum formula for c_{21}: $k = 1$ gives $a_{21}b_{11} = 12 \times 7$ (the Nifty term), $k = 2$ gives $a_{22}b_{21} = 7 \times 3$ (the bond term). One term per fund.

Which shapes can be multiplied

In every dot product, the entries of row i of A (one per fund) are paired with the entries of column j of B (one per fund). For the pairing to work, **the length of a row of A must equal the length of a column of B.**

> **RULE · SHAPES**
> $(m \times n)$ times $(n \times p)$ gives $(m \times p)$. The two inner numbers must match; the two outer numbers give the shape of the answer. Working out C takes $m \times n \times p$ multiplications (mp entries, n multiplications each).

A	B	Allowed?	Result	Portfolio reading
2×2	2×2	yes (2 = 2)	2×2	2 rules, 2 funds, 2 clients
3×3	3×5	yes (3 = 3)	3×5	3 rules, 3 funds, 5 clients
2×3	2×3	**no** (3 ≠ 2)	none	3 funds' coefficients but only 2 holdings per client
1×3	3×1	yes	1×1	a single number: a dot product
3×1	1×3	yes	3×3	a whole matrix: a column times a row (topic 3)

The order matters: XA is a different matrix

Both X and A are 2×2, so XA is also allowed. Entry by entry:

$$XA = \begin{bmatrix} 7 & 4 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix} = \begin{bmatrix} 7 \cdot 1 + 4 \cdot 12 & 7 \cdot 1 + 4 \cdot 7 \\ 3 \cdot 1 + 6 \cdot 12 & 3 \cdot 1 + 6 \cdot 7 \end{bmatrix} = \begin{bmatrix} 55 & 35 \\ 75 & 45 \end{bmatrix} \neq AX.$$

Not only are the numbers different, they mean nothing: $7 \times 1 + 4 \times 12$ mixes client 1's Nifty amount with client 2's Nifty amount times a return. $AB \neq BA$ **in general** (Lesson 2 already warned us).

> **IN ONE LINE**
> Each entry of AB = a row of A dotted with a column of B; the inner sizes must match.

▼ Check yourself: with three funds and four clients, what are the shapes of A, X and AX, and how many multiplications does AX need?

A is 3×3 (3 rules × 3 funds), X is 3×4 (3 funds × 4 clients), AX is 3×4. Multiplications: $3 \times 3 \times 4 = 36$ (12 entries, 3 each).

2. By columns and by rows

Computing entries one at a time is correct, but it hides the structure. Strang's next two views use whole columns and whole rows.

METHOD · COLUMN WAY

Column j of AB is A times column j of B. Since A times a vector is a combination of A's columns (Lesson 1), **every column of AB is a combination of the columns of A.**

1 WHAT WE DO

Get column 1 of AX from client 1's portfolio $(7, 3)$ alone.

WHY

Column 1 of AX uses only column 1 of X: client 2's numbers never enter it (look back at the entry formulas: c_{11} and c_{21} use only 7 and 3).

HOW

$$A\begin{bmatrix}7\\3\end{bmatrix} = 7\begin{bmatrix}1\\12\end{bmatrix} + 3\begin{bmatrix}1\\7\end{bmatrix} = \begin{bmatrix}7\\84\end{bmatrix} + \begin{bmatrix}3\\21\end{bmatrix} = \begin{bmatrix}10\\105\end{bmatrix}.$$

MEANING

7 copies of the Nifty arrow plus 3 copies of the bond arrow: Lesson 1's column picture for client 1.

2 WHAT WE DO

Get column 2 from client 2's portfolio $(4, 6)$.

HOW

$$A\begin{bmatrix}4\\6\end{bmatrix} = 4\begin{bmatrix}1\\12\end{bmatrix} + 6\begin{bmatrix}1\\7\end{bmatrix} = \begin{bmatrix}4\\48\end{bmatrix} + \begin{bmatrix}6\\42\end{bmatrix} = \begin{bmatrix}10\\90\end{bmatrix}.$$

WHAT WE HAVE NOW

Put the two columns side by side: $AX = \begin{bmatrix}10 & 10\\105 & 90\end{bmatrix}$, the same as the entry rule.

METHOD · ROW WAY

Row i of AB is row i of A times B. Since a row times a matrix is a combination of the matrix's rows (Lesson 2), **every row of AB is a combination of the rows of B.**

The rows of X are "₹ in Nifty for each client", $(7, 4)$, and "₹ in bonds for each client", $(3, 6)$.

1 WHAT WE DO — Row 1 of AX: row 1 of A is $(1, 1)$, so take 1 × (row 1 of X) + 1 × (row 2 of X).

HOW — $1\,(7, 4) + 1\,(3, 6) = (7 + 3,\ 4 + 6) = (10, 10)$.

MEANING — Every client's **budget**, in one row: Nifty holdings plus bond holdings.

2 WHAT WE DO — Row 2 of AX: row 2 of A is $(12, 7)$, so take 12 × (row 1 of X) + 7 × (row 2 of X).

HOW — $12\,(7, 4) + 7\,(3, 6) = (84, 48) + (21, 42) = (84 + 21,\ 48 + 42) = (105, 90)$.

MEANING — Every client's **return**: 12 per lakh of Nifty plus 7 per lakh of bonds.

View	One piece of AX is...	Question it answers
Column way	a whole column = one **client**	What does each client end up with?
Row way	a whole row = one **rule**	What is the budget (or return) across all clients?

IN ONE LINE

Columns of AB mix the columns of A; rows of AB mix the rows of B.

▼ Check yourself: a third client holds ₹5 lakh in each fund. What new column appears in AX, by the column way?

$5\,(1, 12) + 5\,(1, 7) = (5, 60) + (5, 35) = (10, 95)$: ₹10 lakh used, ₹95 thousand earned.

3. Columns times rows

A row times a column gives one number. Turn it around: a **column times a row** gives a whole matrix. Each entry is (an entry of the column) × (an entry of the row):

$$\begin{bmatrix}2\\3\\4\end{bmatrix}\begin{bmatrix}1 & 6\end{bmatrix} = \begin{bmatrix}2\times1 & 2\times6\\3\times1 & 3\times6\\4\times1 & 4\times6\end{bmatrix} = \begin{bmatrix}2 & 12\\3 & 18\\4 & 24\end{bmatrix} \qquad \text{(Strang's example)}$$

- Column 1 is $1 \times (2, 3, 4)$ and column 2 is $6 \times (2, 3, 4)$: **every column is a multiple of the column.**
- Row 1 is $2 \times (1, 6)$, row 2 is $3 \times (1, 6)$, row 3 is $4 \times (1, 6)$: **every row is a multiple of the row.**

Strang calls this a "really minimal matrix": all its rows lie on one line, and so do all its columns. Later lessons call it a **rank-one** matrix.

METHOD · COLUMN-TIMES-ROW (SUM OF OUTER PRODUCTS)

$AB = (\text{column 1 of } A)(\text{row 1 of } B) + (\text{column 2 of } A)(\text{row 2 of } B) + \cdots + (\text{column } n \text{ of } A)(\text{row } n \text{ of } B)$. Each term, a column times a row, is called an **outer product.**

1 WHAT WE DO

Multiply column 1 of A, $(1, 12)$ (the Nifty fund), by row 1 of X, $(7, 4)$ (each client's Nifty holding).

HOW

$$\begin{bmatrix} 1 \\ 12 \end{bmatrix} \begin{bmatrix} 7 & 4 \end{bmatrix} = \begin{bmatrix} 1 \times 7 & 1 \times 4 \\ 12 \times 7 & 12 \times 4 \end{bmatrix} = \begin{bmatrix} 7 & 4 \\ 84 & 48 \end{bmatrix}.$$

MEANING

What the **Nifty holdings** contribute: ₹7 lakh and ₹4 lakh of budget, ₹84 and ₹48 thousand of return.

2 WHAT WE DO

Multiply column 2 of A, $(1, 7)$ (bonds), by row 2 of X, $(3, 6)$ (each client's bond holding).

HOW

$$\begin{bmatrix} 1 \\ 7 \end{bmatrix} \begin{bmatrix} 3 & 6 \end{bmatrix} = \begin{bmatrix} 1 \times 3 & 1 \times 6 \\ 7 \times 3 & 7 \times 6 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 21 & 42 \end{bmatrix}.$$

MEANING

What the **bond holdings** contribute.

3 WHAT WE DO

Add the two pieces, entry by entry.

HOW

$$\begin{bmatrix} 7+3 & 4+6 \\ 84+21 & 48+42 \end{bmatrix} = \begin{bmatrix} 10 & 10 \\ 105 & 90 \end{bmatrix}.$$

WHAT WE HAVE NOW

AX again, now split **by fund.**

Why it works: entry $c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j}$ has one term per fund. The first outer product puts exactly the term $a_{i1}b_{1j}$ into every position (i, j), and the second puts exactly $a_{i2}b_{2j}$. Adding them rebuilds every entry.

IN MARKETS: ATTRIBUTION BY SOURCE

"How much of each client's return came from Nifty and how much from bonds?" is called **return attribution.** Here it is literally the column-times-row view: Nifty gave client 1 ₹84 of their ₹105 thousand, bonds gave ₹21.

IN ONE LINE

AB is a sum of column × row pieces, one per inner index (here, one per fund).

4. Block multiplication

Matrices can be cut into smaller rectangles called blocks, and multiplied block by block.

METHOD · BLOCK MULTIPLICATION

Cut A and B into blocks so that the **column cuts of A match the row cuts of B.** Then multiply the blocks exactly as if they were numbers: $\begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix}\begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix}$ has top-left block $A_1B_1 + A_2B_3$, and so on.

Use the full three-fund matrix and three clients (the targets from Lesson 4). Cut A's columns after fund 2, and X's rows after fund 2:

$$A = \left[\begin{array}{cc|c} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{array}\right] = [\,A_{\text{left}} \mid A_{\text{right}}\,], \qquad X = \left[\begin{array}{ccc} 5 & 4 & 6 \\ 3 & 5 & 6 \\ \hline 2 & 1 & 3 \end{array}\right] = \begin{bmatrix} X_{\text{top}} \\ X_{\text{bottom}} \end{bmatrix}$$

A_{left} (3×2) holds the Nifty and bond columns, A_{right} (3×1) the midcap column. X_{top} (2×3) holds every client's Nifty and bond amounts, X_{bottom} (1×3) every client's midcap amount. The cut after 2 columns of A matches the cut after 2 rows of X, so:

$$AX = A_{\text{left}}X_{\text{top}} + A_{\text{right}}X_{\text{bottom}}.$$

1 WHAT WE DO

Compute $A_{\text{left}}X_{\text{top}}$, the Nifty-and-bond part of every client.

HOW

Entry rule, row by row. Row 1 $(1,1)$: $(1\cdot 5 + 1\cdot 3,\ 1\cdot 4 + 1\cdot 5,\ 1\cdot 6 + 1\cdot 6) = (8, 9, 12)$. Row 2 $(12,7)$: $(60 + 21,\ 48 + 35,\ 72 + 42) = (81, 83, 114)$. Row 3 $(1,0)$: $(5 + 0,\ 4 + 0,\ 6 + 0) = (5, 4, 6)$.

WHAT WE HAVE NOW

$A_{\text{left}}X_{\text{top}} = \begin{bmatrix} 8 & 9 & 12 \\ 81 & 83 & 114 \\ 5 & 4 & 6 \end{bmatrix}$.

2 WHAT WE DO — Compute $A_{\text{right}}X_{\text{bottom}}$, the midcap part. It is a column times a row (topic 3).

HOW — $(1, 15, 1.2)^T$ times $(2, 1, 3)$: row 1 $= 1 \times (2, 1, 3) = (2, 1, 3)$; row 2 $= 15 \times (2, 1, 3) = (30, 15, 45)$; row 3 $= 1.2 \times (2, 1, 3) = (2.4, 1.2, 3.6)$.

WHAT WE HAVE NOW — $A_{\text{right}}X_{\text{bottom}} = \begin{bmatrix} 2 & 1 & 3 \\ 30 & 15 & 45 \\ 2.4 & 1.2 & 3.6 \end{bmatrix}$.

3 WHAT WE DO — Add the two blocks.

HOW — Row 1: $(8+2,\ 9+1,\ 12+3) = (10, 10, 15)$. Row 2: $(81 + 30,\ 83 + 15,\ 114 + 45) = (111, 98, 159)$. Row 3: $(5 + 2.4,\ 4 + 1.2,\ 6 + 3.6) = (7.4, 5.2, 9.6)$.

WHAT WE HAVE NOW — The three clients' targets, one client per column:

$$AX = \begin{bmatrix} 10 & 10 & 15 \\ 111 & 98 & 159 \\ 7.4 & 5.2 & 9.6 \end{bmatrix} \begin{matrix} \leftarrow \text{budget} \\ \leftarrow \text{return} \\ \leftarrow \text{exposure} \end{matrix}$$

Large matrices are multiplied in blocks so that each block fits in a computer's fast memory, and so blocks can be multiplied at the same time on different processors.

IN ONE LINE

If the cuts match, blocks multiply like numbers: top-left $= A_1B_1 + A_2B_3$.

▼ **Check yourself (MIT problem 3.1): with $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 0 & 0 \\ 5 & 6 \end{bmatrix}$, compare $AB + AC$ with $A(B + C)$.**

$AB = \begin{bmatrix} 1{\cdot}1+2{\cdot}0 & 1{\cdot}0+2{\cdot}0 \\ 3{\cdot}1+4{\cdot}0 & 3{\cdot}0+4{\cdot}0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 0 \end{bmatrix}$. $AC = \begin{bmatrix} 0+10 & 0+12 \\ 0+20 & 0+24 \end{bmatrix} = \begin{bmatrix} 10 & 12 \\ 20 & 24 \end{bmatrix}$. Sum: $\begin{bmatrix} 11 & 12 \\ 23 & 24 \end{bmatrix}$. And $B + C = \begin{bmatrix} 1 & 0 \\ 5 & 6 \end{bmatrix}$, so $A(B + C) = \begin{bmatrix} 1+10 & 0+12 \\ 3+20 & 0+24 \end{bmatrix} = \begin{bmatrix} 11 & 12 \\ 23 & 24 \end{bmatrix}$. Equal: matrix multiplication is **distributive**, $A(B + C) = AB + AC$.

5. The inverse, and when it exists

With numbers: $5 \times \frac{1}{5} = 1$, so $\frac{1}{5}$ "undoes" multiplying by 5. Every number except 0 has such an inverse; 0 doesn't, because $0 \times$ anything $= 0$, never 1.

For matrices, the role of 1 is played by the identity matrix I. Check that it changes nothing, using the entry rule on our A:

$$IA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix} = \begin{bmatrix} 1\cdot 1 + 0\cdot 12 & 1\cdot 1 + 0\cdot 7 \\ 0\cdot 1 + 1\cdot 12 & 0\cdot 1 + 1\cdot 7 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix} = A.$$

DEFINITION: INVERSE

A square matrix A is **invertible** (also called **non-singular**) if there is a matrix A^{-1} with $A^{-1}A = I$ and $AA^{-1} = I$. Strang: for square matrices, if one of these holds the other holds too, so a left inverse is automatically a right inverse.

Why only square matrices: if A is $m \times n$, then $A^{-1}A = I$ needs A^{-1} to be $n \times m$, and $AA^{-1} = I$ is $m \times m$ while $A^{-1}A$ is $n \times n$. Both can be "the identity" in the same sense only when $m = n$. (Rectangular "one-sided" inverses come in Lesson 31.)

Strang: whether A has an inverse is "the most important question you can ask about the matrix". Start with one that doesn't.

A matrix with no inverse: two identical funds

Lesson 1's two Nifty funds, both 12%: $A = \begin{bmatrix} 1 & 1 \\ 12 & 12 \end{bmatrix}$. Look for a portfolio change x that changes nothing:

$$\begin{bmatrix} 1 & 1 \\ 12 & 12 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1\cdot 1 + 1\cdot(-1) \\ 12\cdot 1 + 12\cdot(-1) \end{bmatrix} = \begin{bmatrix} 1-1 \\ 12-12 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

The trade $x = (1, -1)$: put ₹1 lakh **more** in fund A and ₹1 lakh **less** in fund B. Budget unchanged, return unchanged. Now Strang's argument that no inverse can exist:

1 SUPPOSE — an inverse A^{-1} did exist.

DO — Multiply both sides of $Ax = 0$ on the left by A^{-1}: $A^{-1}(Ax) = A^{-1}0$.

2 LEFT SIDE — Move the parentheses (associative): $A^{-1}(Ax) = (A^{-1}A)x = Ix = x$.

RIGHT SIDE — Any matrix times the zero vector is the zero vector: $A^{-1}0 = 0$.

3 SO — $x = 0$. But $x = (1, -1)$ is **not** zero. The assumption was impossible: **A has no inverse.**

PLAIN WORDS

A sends both the trade $(1, -1)$ and "do nothing" $(0, 0)$ to the same result $(0, 0)$. No matrix can undo that, because it would have to send $(0, 0)$ back to two different places.

RULE · SINGULAR TEST: AX = 0 WITH X ≠ 0

If some **non-zero** vector x has $Ax = 0$, then A has **no inverse** (A is singular). In portfolio terms: a trade that changes nothing means the portfolio for a target is never unique, so it can't be undone. Quick signs (from the recitation): a whole row or column of zeros, or two equal rows or columns. For example, if column j is all zeros then $x =$ (1 in position j, 0 elsewhere) gives $Ax = 0$.

The same in 3 × 3: the balanced fund (half Nifty, half bonds) has column $(1, 9.5, 0.5)$. Try the trade $x = (1, 1, -2)$: buy ₹1 lakh Nifty, buy ₹1 lakh bonds, sell ₹2 lakh balanced.

$$\begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 9.5 \\ 1 & 0 & 0.5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 1+1-2 \\ 12+7-19 \\ 1+0-1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Budget, return and exposure all unchanged: this matrix has no inverse either.

Strang's column reason: every column of a product AX is a combination of A's columns (topic 2). For the two-Nifty matrix both columns are $(1, 12)$, so every combination lies on the line through $(1, 12)$. To get $AX = I$, column 1 of I, $(1, 0)$, would have to lie on that line, and it doesn't. So no X can give I.

IN ONE LINE

$A^{-1}A = I$; if some non-zero trade x has $Ax = 0$, no inverse exists.

6. Gauss–Jordan: 2 × 2

Now a matrix that **does** have an inverse, the two-fund matrix $A = \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}$ (its columns point in different directions). Write the unknown inverse with letters:

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} a & c \\ b & d \end{bmatrix}}_{A^{-1}} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{I}$$

By the column way, this is **two separate systems with the same matrix** A:

- Column 1: $A\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, that is $a + b = 1$ and $12a + 7b = 0$.
- Column 2: $A\begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, that is $c + d = 0$ and $12c + 7d = 1$.

We could solve each by elimination separately. Jordan's idea: carry **both right-hand sides at once**, by augmenting A with the whole identity matrix.

METHOD · GAUSS–JORDAN ELIMINATION

Write $[A \mid I]$. **Gauss:** eliminate downward until the left half is upper triangular. **Jordan:** keep going upward, clearing the entries above each pivot. **Finally:** divide each row by its pivot. When the left half has become I, the right half is A^{-1}. Named after Carl Friedrich Gauss and Wilhelm Jordan (1842–1899), a German geodesist who popularised the upward step.

$$[A \mid I] = \left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 12 & 7 & 0 & 1 \end{array}\right]$$

1

WHAT WE DO — **Gauss:** clear the 12 below the first pivot 1.

WHY — To make the left half upper triangular, as in Lesson 2.

HOW — Multiplier $\ell_{21} = 12/1 = 12$. New row 2 = row 2 − 12 × row 1, entry by entry: $12 - 12 \times 1 = 0$; $7 - 12 \times 1 = -5$; $0 - 12 \times 1 = -12$; $1 - 12 \times 0 = 1$.

WHAT WE HAVE NOW — $\left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & -5 & -12 & 1 \end{array}\right]$. Upper triangular on the left: Gauss alone would stop here.

2

WHAT WE DO — **Jordan:** clear the 1 **above** the second pivot -5.

WHY — To turn the left half into the identity, every entry above the diagonal must become 0 too.

HOW — Multiplier $= 1/(-5) = -0.2$, so new row 1 = row 1 − (−0.2) × row 2 = row 1 + 0.2 × row 2: $1 + 0.2 \times 0 = 1$; $1 + 0.2 \times (-5) = 0$; $1 + 0.2 \times (-12) = 1 - 2.4 = -1.4$; $0 + 0.2 \times 1 = 0.2$.

WHAT WE HAVE NOW — $\left[\begin{array}{cc|cc} 1 & 0 & -1.4 & 0.2 \\ 0 & -5 & -12 & 1 \end{array}\right]$. The left half is diagonal.

3

WHAT WE DO — Divide row 2 by its pivot -5.

HOW $0/(-5) = 0;\ -5/(-5) = 1;\ -12/(-5) = 2.4;\ 1/(-5) = -0.2.$
(Row 1's pivot is already 1.)

WHAT WE HAVE NOW $\left[\begin{array}{cc|cc} 1 & 0 & -1.4 & 0.2 \\ 0 & 1 & 2.4 & -0.2 \end{array}\right] = [\,I \mid A^{-1}\,].$

$$A^{-1} = \begin{bmatrix} -1.4 & 0.2 \\ 2.4 & -0.2 \end{bmatrix}$$

Full check, both orders. $A^{-1}A$, entry by entry:

$$\begin{bmatrix} -1.4 \cdot 1 + 0.2 \cdot 12 & -1.4 \cdot 1 + 0.2 \cdot 7 \\ 2.4 \cdot 1 - 0.2 \cdot 12 & 2.4 \cdot 1 - 0.2 \cdot 7 \end{bmatrix} = \begin{bmatrix} -1.4 + 2.4 & -1.4 + 1.4 \\ 2.4 - 2.4 & 2.4 - 1.4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \checkmark$$

$$AA^{-1} = \begin{bmatrix} 1 \cdot (-1.4) + 1 \cdot 2.4 & 1 \cdot 0.2 + 1 \cdot (-0.2) \\ 12 \cdot (-1.4) + 7 \cdot 2.4 & 12 \cdot 0.2 + 7 \cdot (-0.2) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -16.8 + 16.8 & 2.4 - 1.4 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

What A^{-1} tells us

A and A^{-1} answer opposite questions:

Matrix	You give it…	It gives you…	Question it answers
A	a portfolio (x, y): ₹ lakh in Nifty and bonds	its results (B, R): budget and return	"What does this portfolio do?"
A^{-1}	the results you **want**: (B, R)	the portfolio (x, y) that delivers them	"What portfolio hits my target?"

So A^{-1} is a ready-made answer machine: $\begin{bmatrix} x \\ y \end{bmatrix} = A^{-1} \begin{bmatrix} B \\ R \end{bmatrix} = \begin{bmatrix} -1.4 & 0.2 \\ 2.4 & -0.2 \end{bmatrix} \begin{bmatrix} B \\ R \end{bmatrix}$. Its four numbers can be read two ways, by columns and by rows.

Reading A^{-1} by columns: what one extra unit of a target costs

By the column way, $A^{-1}(B, R) = B \times (\text{column } 1) + R \times (\text{column } 2)$. So each column is **the change in the portfolio needed for one extra unit of one target, with the other target held fixed.** First, one fact does all the work:

> THE KEY FACT
>
> Every lakh you switch **from bonds to Nifty** keeps the budget the same and adds $12 - 7 =$ **₹5 thousand** of return. Every lakh switched the other way removes ₹5 thousand. (This 5 is the same 5 that sits in $ad - bc = -5$.)

1 COLUMN 2

$(0.2, -0.2)$: the change for **+₹1 thousand of return, same budget.**

WHY THESE NUMBERS

The budget must not move, so the only option is to switch money between the funds. Each lakh switched from bonds to Nifty adds ₹5 thousand, so ₹1 thousand needs $1/5 = 0.2$ lakh switched: Nifty $+0.2$, bonds -0.2.

CHECK

Budget: $0.2 - 0.2 = 0$ ✓. Return: $12(0.2) + 7(-0.2) = 2.4 - 1.4 = 1$ ✓.

IN ₹

Move ₹20,000 from the bond fund to the Nifty fund, and the yearly return rises by ₹1,000.

2 COLUMN 1

$(-1.4, 2.4)$: the change for **+₹1 lakh of budget, same return.**

WHY THESE NUMBERS

Think in two moves. First add the new ₹1 lakh to bonds: budget $+1$, but return $+7$ thousand, which we don't want. Then cancel that ₹7 thousand by switching money from Nifty back to bonds: at ₹5 thousand per lakh that is $7/5 = 1.4$ lakh. Total: bonds $1 + 1.4 = 2.4$, Nifty -1.4.

CHECK

Budget: $-1.4 + 2.4 = 1$ ✓. Return: $12(-1.4) + 7(2.4) = -16.8 + 16.8 = 0$ ✓.

IN PLAIN WORDS

Extra money with no extra return can only be absorbed by tilting towards the low-return fund: the new ₹1 lakh goes into bonds, and some Nifty is swapped into bonds too.

Building any target from the two columns

Because $x = B \times (\text{column 1}) + R \times (\text{column 2})$, every target is just "budget × column 1 + return × column 2":

1 TARGET

₹10 lakh, ₹105 thousand return (the running example).

HOW

$10 \times (-1.4, 2.4) + 105 \times (0.2, -0.2) = (-14, 24) + (21, -21) = (-14 + 21,\ 24 - 21) = (7, 3)$.

ANSWER

₹7 lakh Nifty, ₹3 lakh bonds, the answer from Lessons 1 and 2.

2 TARGET

₹10 lakh at 10%, i.e. ₹100 thousand.

HOW

$10 \times (-1.4, 2.4) + 100 \times (0.2, -0.2) = (-14, 24) + (20, -20) = (6, 4)$.

ANSWER ₹6 lakh Nifty, ₹4 lakh bonds. Check: $6+4=10$ ✓, $72+28=100$ ✓.

3 TARGET ₹10 lakh at 12%, i.e. ₹120 thousand.

HOW $(-14,24)+120\times(0.2,-0.2)=(-14+24,\ 24-24)=(10,0)$.

ANSWER Everything in Nifty, which makes sense: only Nifty earns 12%.

Reading A^{-1} by rows: one formula per fund

Each row of A^{-1} gives one fund's amount directly from the targets:

$$\text{Nifty} = -1.4\,B + 0.2\,R, \qquad \text{Bonds} = 2.4\,B - 0.2\,R.$$

A client with ₹12 lakh who wants ₹120 thousand: Nifty $= -1.4(12)+0.2(120) = -16.8+24 = 7.2$ and bonds $= 2.4(12)-0.2(120) = 28.8-24 = 4.8$. Check: $7.2+4.8=12$ ✓, $12(7.2)+7(4.8) = 86.4+33.6 = 120$ ✓.

IN ONE LINE

A^{-1} turns "what I want" into "what to hold". Its columns are the price of one extra unit of each target; its rows are one formula per fund.

IN ONE LINE

$[A \mid I] \to [I \mid A^{-1}]$: eliminate down (Gauss), up (Jordan), then divide by the pivots.

▼ Check yourself (Strang's example): invert $\left[\begin{smallmatrix}1&3\\2&7\end{smallmatrix}\right]$ by Gauss–Jordan.

Gauss: $\ell_{21}=2$; row 2 − 2 × row 1: $(2-2,\ 7-6 \mid 0-2,\ 1-0) = (0,1 \mid -2,1)$. **Jordan:** row 1 − 3 × row 2: $(1-0,\ 3-3 \mid 1+6,\ 0-3) = (1,0 \mid 7,-3)$. Pivots are already 1. So $A^{-1} = \left[\begin{smallmatrix}7&-3\\-2&1\end{smallmatrix}\right]$. Check row 1 of $A^{-1}A$: $(7-6,\ 21-21) = (1,0)$ ✓.

7. Gauss–Jordan: 3 × 3, and why it works

The three-fund portfolio. Start from $[A \mid I]$ (rows: budget, return, exposure):

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 12 & 7 & 15 & 0 & 1 & 0 \\ 1 & 0 & 1.2 & 0 & 0 & 1 \end{array}\right]$$

Gauss: downward

1 WHAT WE DO — Clear the 12 in row 2, column 1.

HOW — $\ell_{21} = 12/1 = 12$. Row 2 − 12 × row 1, all six entries: $12 - 12 = 0$; $7 - 12 = -5$; $15 - 12 = 3$; $0 - 12 = -12$; $1 - 0 = 1$; $0 - 0 = 0$.

WHAT WE HAVE NOW — Row 2 $= (0, -5, 3 \mid -12, 1, 0)$.

2 WHAT WE DO — Clear the 1 in row 3, column 1.

HOW — $\ell_{31} = 1/1 = 1$. Row 3 − 1 × row 1: $1 - 1 = 0$; $0 - 1 = -1$; $1.2 - 1 = 0.2$; $0 - 1 = -1$; $0 - 0 = 0$; $1 - 0 = 1$.

WHAT WE HAVE NOW — Row 3 $= (0, -1, 0.2 \mid -1, 0, 1)$.

3 WHAT WE DO — Clear the -1 in row 3, column 2, using the second pivot -5.

HOW — $\ell_{32} = (-1)/(-5) = 0.2$. Row 3 − 0.2 × row 2: $0 - 0 = 0$; $-1 - 0.2(-5) = -1 + 1 = 0$; $0.2 - 0.2(3) = 0.2 - 0.6 = -0.4$; $-1 - 0.2(-12) = -1 + 2.4 = 1.4$; $0 - 0.2(1) = -0.2$; $1 - 0.2(0) = 1$.

WHAT WE HAVE NOW — Row 3 $= (0, 0, -0.4 \mid 1.4, -0.2, 1)$, and the left half is upper triangular:

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -5 & 3 & -12 & 1 & 0 \\ 0 & 0 & -0.4 & 1.4 & -0.2 & 1 \end{array}\right]$$

Look at the right half: it is exactly $E = E_{32}E_{31}E_{21}$ from Lesson 2, the messy 1.4 included. The identity has quietly recorded every row operation.

Jordan: upward

1 WHAT WE DO — Clear the 3 above the third pivot (row 2, column 3).

HOW — Multiplier $= 3/(-0.4) = -7.5$, so row 2 − (−7.5) × row 3 = row 2 + 7.5 × row 3: $0 + 0 = 0$; $-5 + 0 = -5$; $3 + 7.5(-0.4) = 3 - 3 = 0$

; $-12 + 7.5(1.4) = -12 + 10.5 = -1.5$; $1 + 7.5(-0.2) = 1 - 1.5 = -0.5$; $0 + 7.5(1) = 7.5$.

WHAT WE HAVE NOW — Row 2 $= (0, -5, 0 \mid -1.5, -0.5, 7.5)$.

2 WHAT WE DO — Clear the 1 above the third pivot in row 1.

HOW — Multiplier $= 1/(-0.4) = -2.5$, so row 1 + 2.5 × row 3: $1 + 0 = 1$; $1 + 0 = 1$; $1 + 2.5(-0.4) = 1 - 1 = 0$; $1 + 2.5(1.4) = 1 + 3.5 = 4.5$; $0 + 2.5(-0.2) = -0.5$; $0 + 2.5(1) = 2.5$.

WHAT WE HAVE NOW — Row 1 $= (1, 1, 0 \mid 4.5, -0.5, 2.5)$.

3 WHAT WE DO — Clear the 1 above the second pivot -5 (row 1, column 2).

HOW — Multiplier $= 1/(-5) = -0.2$, so row 1 + 0.2 × row 2: $1 + 0 = 1$; $1 + 0.2(-5) = 0$; $0 + 0 = 0$; $4.5 + 0.2(-1.5) = 4.5 - 0.3 = 4.2$; $-0.5 + 0.2(-0.5) = -0.5 - 0.1 = -0.6$; $2.5 + 0.2(7.5) = 2.5 + 1.5 = 4$.

WHAT WE HAVE NOW — Row 1 $= (1, 0, 0 \mid 4.2, -0.6, 4)$. The left half is diagonal: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & -0.4 \end{bmatrix}$.

Divide by the pivots

1 WHAT WE DO — Row 2 ÷ (-5) and row 3 ÷ (-0.4) (row 1's pivot is 1).

HOW — Row 2: $-1.5/(-5) = 0.3$; $-0.5/(-5) = 0.1$; $7.5/(-5) = -1.5$.
Row 3: $1.4/(-0.4) = -3.5$; $-0.2/(-0.4) = 0.5$; $1/(-0.4) = -2.5$.

WHAT WE HAVE NOW — $[\,I \mid A^{-1}\,]$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 4.2 & -0.6 & 4 \\ 0 & 1 & 0 & 0.3 & 0.1 & -1.5 \\ 0 & 0 & 1 & -3.5 & 0.5 & -2.5 \end{array}\right] \qquad A^{-1} = \begin{bmatrix} 4.2 & -0.6 & 4 \\ 0.3 & 0.1 & -1.5 \\ -3.5 & 0.5 & -2.5 \end{bmatrix}$$

Full check: all nine entries of $A^{-1}A$

Each entry is (row of A^{-1}) · (column of A). The columns of A are $(1, 12, 1)$, $(1, 7, 0)$, $(1, 15, 1.2)$.

Row of A^{-1}	· column 1 $(1, 12, 1)$	· column 2 $(1, 7, 0)$	· column 3 $(1, 15, 1.2)$
$(4.2, -0.6, 4)$	$4.2 - 7.2 + 4 = 1$	$4.2 - 4.2 + 0 = 0$	$4.2 - 9 + 4.8 = 0$
$(0.3, 0.1, -1.5)$	$0.3 + 1.2 - 1.5 = 0$	$0.3 + 0.7 + 0 = 1$	$0.3 + 1.5 - 1.8 = 0$
$(-3.5, 0.5, -2.5)$	$-3.5 + 6 - 2.5 = 0$	$-3.5 + 3.5 + 0 = 0$	$-3.5 + 7.5 - 3 = 1$

1s on the diagonal, 0s elsewhere: $A^{-1}A = I$ ✓.

WHY GAUSS–JORDAN WORKS

Every row operation is a multiplication on the left by an elimination matrix (Lesson 2). Call the product of all of them E. Applying E to the long matrix acts on both halves at once (block multiplication, topic 4):

$$E\,[\,A \mid I\,] = [\,EA \mid EI\,] = [\,EA \mid E\,].$$

We stopped exactly when the left half became I, so $EA = I$. That says E undoes A: $E = A^{-1}$. And the right half is E itself. So the right half **has** to be A^{-1}. (Strang: "It tells us what E has to be.")

IN ONE LINE

The identity on the right records every row operation; when the left becomes I, the right is A^{-1}.

8. Using the inverse

Every client at once: X = A⁻¹B

The clients want targets B, and we need portfolios X with $AX = B$. Multiply both sides on the left by A^{-1}: $A^{-1}AX = A^{-1}B$, and since $A^{-1}A = I$, this is $X = A^{-1}B$. Each column of B (one client's target) gives one column of X (that client's portfolio). Row by row of A^{-1}:

Client (target)	Nifty = $(4.2, -0.6, 4) \cdot b$	Bonds = $(0.3, 0.1, -1.5) \cdot b$	Midcap = $(-3.5, 0.5, -2.5) \cdot b$
1: $(10, 111, 7.4)$	$42 - 66.6 + 29.6 = 5$	$3 + 11.1 - 11.1 = 3$	$-35 + 55.5 - 18.5 = 2$
2: $(10, 98, 5.2)$	$42 - 58.8 + 20.8 = 4$	$3 + 9.8 - 7.8 = 5$	$-35 + 49 - 13 = 1$
3: $(15, 159, 9.6)$	$63 - 95.4 + 38.4 = 6$	$4.5 + 15.9 - 14.4 = 6$	$-52.5 + 79.5 - 24 = 3$

So $X = \begin{bmatrix} 5 & 4 & 6 \\ 3 & 5 & 6 \\ 2 & 1 & 3 \end{bmatrix}$: client 1 holds $(5, 3, 2)$, client 2 holds $(4, 5, 1)$, client 3 holds $(6, 6, 3)$, all in one product and matching Lessons 1–4.

The columns of A^{-1} are building-block trades

Read $AA^{-1} = I$ by the column way: A times column j of A^{-1} = column j of I. So **column j of A^{-1} is the trade that changes rule j by exactly 1 and leaves every other rule unchanged.** Each one checked in full:

Column of A^{-1}	Trade (Nifty, bonds, midcap, ₹ lakh)	Budget change	Return change	Exposure change
1	$(4.2,\ 0.3,\ -3.5)$	$4.2 + 0.3 - 3.5 = 1$	$50.4 + 2.1 - 52.5 = 0$	$4.2 + 0 - 4.2 = 0$
2	$(-0.6,\ 0.1,\ 0.5)$	$-0.6 + 0.1 + 0.5 = 0$	$-7.2 + 0.7 + 7.5 = 1$	$-0.6 + 0 + 0.6 = 0$
3	$(4,\ -1.5,\ -2.5)$	$4 - 1.5 - 2.5 = 0$	$48 - 10.5 - 37.5 = 0$	$4 + 0 - 3 = 1$

Use one: a client wants **₹5 thousand more return, nothing else changed.** Take 5 × column 2: $5 \times (-0.6, 0.1, 0.5) = (-3, 0.5, 2.5)$. That means move ₹3 lakh out of Nifty, put ₹0.5 lakh into bonds and ₹2.5 lakh into midcap. Check: budget $-3 + 0.5 + 2.5 = 0$; return $-36 + 3.5 + 37.5 = 5$; exposure $-3 + 0 + 3 = 0$ ✓. Desks call such columns **sensitivities**: how far each position must move per unit change in one target.

The 2 × 2 shortcut, and why it works

RULE · 2 × 2 INVERSE FORMULA

$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \dfrac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$, valid when $ad - bc \neq 0$: swap the diagonal, negate the off-diagonal, divide by $ad - bc$.

Why it works: multiply the bracket by the original matrix, entry by entry:

$$\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} da - bc & db - bd \\ -ca + ac & -cb + ad \end{bmatrix} = \begin{bmatrix} ad - bc & 0 \\ 0 & ad - bc \end{bmatrix}.$$

Dividing by $ad - bc$ turns that into I. If $ad - bc = 0$ the division is impossible: no inverse. For the two funds, $a = 1, b = 1, c = 12, d = 7$: $ad - bc = 7 - 12 = -5$, and

$$A^{-1} = \frac{1}{-5} \begin{bmatrix} 7 & -1 \\ -12 & 1 \end{bmatrix} = \begin{bmatrix} -1.4 & 0.2 \\ 2.4 & -0.2 \end{bmatrix},$$

the same as Gauss–Jordan. For the two identical Nifty funds, $ad - bc = 12 - 12 = 0$: no inverse, as we found. The number $ad - bc$ is the *determinant*, which gets whole lessons later.

Lesson 1's formula falls out: for budget B and return R, $x = A^{-1}(B, R)$ gives Nifty $= -1.4B + 0.2R = (R - 7B)/5$ and bonds $= 2.4B - 0.2R = (12B - R)/5$, the formula derived by substitution in Lesson 1.

WATCH OUT: YOU RARELY NEED A^{-1} ITSELF

To solve $Ax = b$, don't compute A^{-1} and then multiply. Factor $A = LU$ once and do two triangular solves (Lesson 4): fewer operations and smaller rounding errors. Compute A^{-1} when you want its entries for their own sake, such as the sensitivities above.

IN ONE LINE

$X = A^{-1}B$ solves every target at once, and the columns of A^{-1} are the one-target-at-a-time trades.